

Biological Model

Dan Rivenco

June 2026

1 Purpose of the Biological Model

The purpose of the biological model is to describe the relationship between three biological processes in the system: the feeding of the mussel, the production of biological waste, and the use of this waste as a nutrient source for algae growth. This is used for a simple engineering model that helps decide when the algae pump and the waste pump should be activated.

The system consists of two connected water containers: one containing algae and one containing a mussel. The algae pump transfers algae-rich water from the algae container to the mussel container. The mussel filters algae from the water as food and produces biological waste. The waste pump then transfers nutrient-rich water from the mussel container back to the algae container. In this way, the model describes a simple biological recycling loop:

Algae $\rightarrow$ Mussel feeding $\rightarrow$ Waste production $\rightarrow$ Algae nutrient input

2 Model Variables

The main variables used in the biological model are shown in Table 1.

Symbol	Description	Unit
$A_m(t)$	Algae concentration in the mussel container	cells/mL or OD units
$A_a(t)$	Algae concentration in the algae container	cells/mL or OD units
$W_m(t)$	Waste concentration in the mussel container	relative units
$W_a(t)$	Waste/nutrient concentration in the algae container	relative units
N_m	Number of mussels	mussels
k_f	Mussel feeding/filtration coefficient	h^{-1}
μ	Natural algae growth rate	h^{-1}
y	Waste yield from mussel feeding	waste/algae
η	Effect of waste nutrients on algae growth	h^{-1}
$u_a(t)$	Algae pump state	0 or 1
$u_w(t)$	Waste pump state	0 or 1
α_a	Normalized algae pump rate	h^{-1}
α_w	Normalized waste pump rate	h^{-1}

Table 1: Variables used in the biological model.

The pump states are defined as:

$$u_a(t) = \begin{cases} 1, & \text{algae pump ON} \\ 0, & \text{algae pump OFF} \end{cases}$$

$$u_w(t) = \begin{cases} 1, & \text{waste pump ON} \\ 0, & \text{waste pump OFF} \end{cases}$$

The normalized pump rates α_a and α_w are used because the exact water volumes are unknown. In a complete system, these would be defined as:

$$\alpha_a = \frac{Q_a}{V_m}$$

$$\alpha_w = \frac{Q_w}{V_m}$$

where Q_a is the algae pump flow rate, Q_w is the waste pump flow rate, and V_m is the mussel container volume. Since V_m is not known in this prototype, α_a and α_w are treated as experimentally adjustable parameters.

3 Mussel Feeding Model

The mussel feeds by filtering algae from the water. Therefore, the algae concentration in the mussel container decreases over time due to mussel feeding. For one mussel, the feeding term is modelled as:

$$R_f = k_f A_m$$

where R_f is the rate at which algae are removed by the mussel and k_f is the filtration coefficient.

Because only one mussel is used in the experiment:

$$N_m = 1$$

The general feeding model for multiple mussels would be:

$$R_f = k_f N_m A_m$$

but for this experiment it becomes:

$$R_f = k_f A_m$$

Based on the previous experimental data, the algae concentration decreased during the mussel feeding test. A representative value for the filtration coefficient is:

$$k_f \approx 0.35 \text{ h}^{-1}$$

This means that the mussel removes a significant fraction of the available algae over time. The model therefore assumes that the mussel tank must be periodically supplied with algae using the algae pump.

4 Algae Growth Model

The algae concentration in the algae container increases naturally due to biological growth. A simple exponential growth model is used:

$$\frac{dA_a}{dt} = \mu A_a$$

where μ is the algae growth rate. From the previous algae growth data, the estimated growth rate was:

$$\mu \approx 0.0351 \text{ h}^{-1}$$

This corresponds to a doubling time of:

$$T_d = \frac{\ln(2)}{\mu}$$

$$T_d = \frac{\ln(2)}{0.0351} \approx 19.7 \text{ h}$$

This shows that algae growth is relatively slow compared with the mussel feeding process. Therefore, the algae pump is necessary to actively feed the mussel container during the experiment.

4.1 Waste Production Model

The mussel produces biological waste as a result of feeding. The waste production is assumed to be proportional to the amount of algae consumed by the mussel. Therefore, the waste production rate is written as:

$$R_w = yR_f$$

Substituting the mussel feeding rate gives:

$$R_w = yk_f A_m$$

where y is the waste yield coefficient. This coefficient represents how much biological waste is produced per unit of algae consumed.

The waste concentration in the mussel container is then modelled as:

$$\frac{dW_m}{dt} = yk_f A_m - u_w \alpha_w W_m$$

The first term, $yk_f A_m$, represents waste produced by the mussel. The second term, $u_w \alpha_w W_m$, represents waste removed from the mussel container when the waste pump is activated.

5 Waste as Nutrient Input for Algae

The biological waste removed from the mussel container is transferred back to the algae container. In the model, this waste is treated as a nutrient input that can support algae growth. Therefore, the algae growth model is extended with a waste-dependent growth term:

$$\frac{dA_a}{dt} = \mu A_a + \eta W_a A_a - u_a \alpha_a A_a$$

The term μA_a represents natural algae growth. The term $\eta W_a A_a$ represents the additional algae growth supported by nutrients from mussel waste. The term $u_a \alpha_a A_a$ represents algae removed from the algae container when the algae pump sends algae-rich water to the mussel container.

The nutrient concentration in the algae container is modelled as:

$$\frac{dW_a}{dt} = u_w \alpha_w W_m - \eta W_a A_a$$

The first term represents waste transferred from the mussel container to the algae container. The second term represents nutrient uptake by algae during growth.

6 Final Coupled Biological Model

The final biological model combines mussel feeding, waste production, algae growth, and pump action. For one mussel, the model is:

$$\frac{dA_m}{dt} = u_a \alpha_a (A_a - A_m) - k_f A_m$$

$$\frac{dW_m}{dt} = y k_f A_m - u_w \alpha_w W_m$$

$$\frac{dA_a}{dt} = \mu A_a + \eta W_a A_a - u_a \alpha_a A_a$$

$$\frac{dW_a}{dt} = u_w \alpha_w W_m - \eta W_a A_a$$

These four equations describe the biological cycle of the prototype.

The first equation describes the algae concentration in the mussel container. Algae are added by the algae pump and removed by mussel feeding.

The second equation describes waste accumulation in the mussel container. Waste is produced when the mussel feeds and removed when the waste pump is activated.

The third equation describes algae growth in the algae container. Algae grow naturally, receive additional support from nutrients in mussel waste, and are removed when pumped into the mussel container.

The fourth equation describes the nutrient/waste concentration in the algae container. Waste enters from the mussel container and is gradually used by the algae.

7 Pump Control Strategy

The model can be used to define a simple control strategy for the two pumps.

The algae pump should be activated when the algae concentration in the mussel container is below the desired feeding level:

$$u_a = \begin{cases} 1, & A_m < A_{\text{low}} \\ 0, & A_m \geq A_{\text{target}} \end{cases}$$

where A_{low} is the minimum acceptable algae concentration and A_{target} is the desired feeding concentration.

The waste pump should be activated after feeding or at fixed time intervals to prevent waste accumulation in the mussel container:

$$u_w = \begin{cases} 1, & W_m > W_{\text{max}} \\ 0, & W_m \leq W_{\text{max}} \end{cases}$$

In the practical prototype, W_m is not measured directly. Therefore, the waste pump can instead be activated using a time-based rule:

$$u_w = 1 \quad \text{for a short period after algae feeding}$$

This means that after the algae pump feeds the mussel, the system waits for a short time and then runs the waste pump briefly. This removes nutrient-rich water from the mussel container and transfers it back to the algae container.

8 Interpretation of the Model

The model shows that the system works as a biological feedback loop. The algae pump provides food to the mussel. The mussel consumes algae and produces waste. The waste pump removes this waste and returns it to the algae container, where it can act as a nutrient source for algae growth.

A key result is that the mussel feeding rate is faster than the algae growth rate. Using the estimated values:

$$k_f \approx 0.35 \text{ h}^{-1}$$

$$\mu \approx 0.0351 \text{ h}^{-1}$$

The ratio is:

$$\frac{k_f}{\mu} = \frac{0.35}{0.0351} \approx 10$$

This means that the mussel removes algae approximately ten times faster than the algae naturally grow. Therefore, the system cannot rely only on algae growth during the experiment. The algae pump is required to maintain enough food in the mussel container.

9 Limitations of the Model

The model is simplified because the exact water volumes are unknown and the experiment uses only one mussel. The waste concentration is also not directly measured, so the waste pump is controlled by timing rather than a real waste sensor. The coefficients γ and η are not known exactly and would need to be determined experimentally in a more complete biological study.

However, the model is still useful for the prototype because it explains the main biological relationships and provides a basis for pump control. It connects mussel feeding, waste production, and algae growth in a single mathematical framework.

10 Conclusion

The biological model describes the relationship between algae feeding, mussel filtration, waste production, and algae nutrient recycling. The algae pump controls the delivery of food to the mussel. The mussel consumes algae and produces waste. The waste pump transfers this waste back to the algae container, where it can support algae growth.

Because the system is tested with one mussel and unknown container volumes, the model uses concentration-based equations and normalized pump rates. This makes the model suitable for the current prototype while still allowing future improvements when the exact volumes, pump flow rates, and biological coefficients are measured.